

The Comparative Effectiveness of Portable Video Modeling vs. Traditional Video Modeling Interventions with Children with Autism Spectrum Disorders

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Abstract Newly developed technologies are being incorporated into treatments for children with autism but there is only limited research examining how this affects child outcomes. In the present study, a multiple baseline design across children and an adapted alternating treatments design within child were used to compare the effectiveness of video modeling interventions implemented on the iPad® to video modeling interventions implemented on the traditionally used television. Two targets were selected for each child and behaviors were randomly assigned to a treatment condition. Target behaviors were acquired and these behavior changes were found to be socially valid across treatment conditions. Four of the children required slightly more treatment sessions to acquire behaviors targeted on the iPad®. Generalization and maintenance were slightly higher following video modeling on the iPad®, but differences were inconsistent and unlikely to result in meaningfully different outcomes. These findings indicate that video modeling on the iPad® does not result in improved child progress compared to traditional video modeling procedures. Additional research is needed to further examine the impact of using portable handheld devices when implementing video modeling interventions.

Keywords Autism · Video modeling · Social-communication skills · Technology

Video modeling is commonly used to treat children with Autism Spectrum Disorder (American Psychiatric Association 2013). With video modeling, the child watches videotaped clips of the targeted behavior(s) being modeled. The child is then provided with opportunities to engage in the modeled behavior(s). This continues until the targeted behaviors are consistently performed by the child (Bellini and Akullian 2007). Many studies demonstrate that video modeling interventions implemented on

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the traditionally used television or computer screen successfully promote a variety of behaviors. Past research has shown that video modeling has increased mands (Plavnick and Ferren 2011), conversational speech and variation in conversational speech (Charlop et al. 2009; Charlop and Milstein 1989; Sherer, Pierce, Paredes, Kisacky, Ingersoll, and Schreibman 2001), solitary and reciprocal pretend play actions and verbalizations (MacDonald, Clark, Garrigan, and Vangala 2005; MacDonald et al. 2009), adaptive skills (e.g., setting the table, mailing a letter; Shipley-Benamou, Lutzker, & Taubman 2002), and vocational skills (Allen, Wallace, Renes, Bowen, and Burke 2010).

Recently developed portable handheld technologies (e.g., the iPad®, Android®, and Kindle Fire®) are being incorporated into video modeling procedures. Implementing the interventions on portable handheld devices is thought to have several advantages. First, the portability of the device enables the intervention to be shown in the setting that is most appropriate to the skill (e.g., in the classroom, at the park). Doing so would be advantageous as children with ASD often fail to generalize skills learned in clinical settings to other places, people, and stimuli (e.g., Schreibman 2000; Stokes and Baer 1977). Second, it has been suggested that children with ASD prefer portable handheld devices and that their use increases the children's motivation to participate in and respond to treatment (Goldsmith and LeBlanc 2004; Kagohara, Sigafos, Achmadi, O'Reilly, and Lancioni 2012). Finally, it is possible that implementing interventions on common and popular devices increases the social validity and decreases the stigma of the intervention procedures (Goldsmith and LeBlanc 2004; Shane et al. 2011).

Several studies provide preliminary evidence that implementing video modeling interventions on portable handheld devices effectively facilitates skill acquisition, but methodological limitations temper the strength of these findings. One study found that video modeling on an iPod® increased the independent transitions of children with ASD throughout the school day when prompting procedures were also in place (Cihak, Fahrenkrog, Ayres, and Smith 2010). Using what was described as a video self-modeling intervention on an iPad®, teaching staff have also increased the correct responding demonstrated by an adolescent with ASD (Hart and Whalon 2012). These studies support the use of video modeling on portable handheld technologies as part of a treatment package. In another study, two children with ASD who watched point-of-view video modeling clips on an iPad® improved their spell-check performance (Kagohara et al. 2012).

Two studies directly compared the effectiveness of video modeling interventions implemented on different devices. Using an adapted alternating treatments design, one study measured the fine motor tasks demonstrated by four boys with ASD (Mechling and Youhouse 2012). After three baseline probes measuring pretreatment demonstration, one group of tasks was targeted with video modeling on a laptop and another group of tasks was targeted with video modeling on a PDA. Three boys performed similarly on both groups of tasks and the fourth performed slightly better on the tasks viewed on the PDA over six treatment sessions. In contrast, a replication study conducted with 19- to 21-year-olds with ASD found that all four participants exhibited consistently better demonstration of the targeted fine motor skills following video modeling on the bigger laptop screen (Mechling and Ayres 2012). In fact, the six or seven video modeling presentations on the PDA did not enable the adults to achieve

criterion level responding. The mixed findings presented in these two studies highlight the need for additional research in this area.

The present study was designed to contribute to the literature by using an experimental design to directly compare the effectiveness of video modeling on the television to video modeling on a portable device, the iPad®, across children, behaviors, and settings and examine social validity of outcomes.

Materials and Method

Participants

Participants were five children with ASD (see Table 1). The children were 5 to 12 years old, had independent diagnoses of an ASD, attended a weekly after-school behavioral treatment program, attended to short video clips, and had access to a television and iPad® in their homes. Two children had prior experience with video modeling interventions and both had more exposure to video modeling on portable handheld devices.

According to parent report, Logan rarely initiated peer interactions or demonstrated the language and play skills required to participate in peer activities. His mother said that he demonstrated interest in preferred stimuli and manipulated items, but seldom engaged in more advanced pretend or socio-dramatic play.

Despite her age-appropriate vocabulary scores, Anna's parents indicated that she exhibited inappropriate speech (e.g., repeatedly saying a word or phrase, saying phrases that did not make sense). Anna spoke in complete sentences of three or more words when prompted but she only occasionally did so independently.

According to his mother, Brian exhibited interest in others and usually responded to social initiations. She said that he often spoke in 2 to 3 word phrases and mixed up the ordering of words. She said that he pointed to stimuli or asked "What" questions (e.g., "What's that?"), but that he rarely asked other types of questions (e.g., "Who," "Where," "When," etc.). His mother also felt that Brian tended to be more active and restless than other boys his age and had difficulty attending to instruction for extended

Table 1 Child characteristics

Child	Ethnicity	Chron. Age (at start of baseline)	Vineland-II Adaptive Level	Vineland-II Comm. and Soc. Percentile	PPVT-IV Age-Equiv.	EVT-II Age-Equiv.
Logan	Caucasian	11–3	Low	0.5 1	7–1	7–5
Anna	Chinese	5–0	Adequate	75 10	4–9	5–1
Brian	Caucasian	8–4	Mod. Low	6 7	3–9	6–3
Jason	Vietnamese	12–1	Mod. Low	2 7	6–8	6–10
Riley	Chinese	7–2	Mod. Low	21 7	5–5	5–9

periods of time. She said that he sometimes demonstrated extreme noncompliant behaviors (e.g., crying, screaming, spitting, elopement).

According to his mother, Jason occasionally joined peer activities (e.g., games, art projects), but seldom initiated conversations. She said that he sometimes responded to social initiations, but rarely attempted to sustain interactions by elaborating on answers or asking his interaction partner a question. He demonstrated echolalic speech and did not always make eye contact. His mother reported that Jason often ignored those around him and had trouble attending to instruction for prolonged periods of time.

Riley's mother reported that he occasionally engaged in interactions with others, but rarely sustained interactions without prompting. She also said that his speech was sometimes inaudible. According to his mother, Riley occasionally played simple board games (e.g., Candyland), but did not consistently engage in other types of interactive play. His mother reported that Riley tended to have difficulty attending to instruction.

Target Behaviors, Assignment to Condition, and Measures

Two behaviors were selected for each child (see Table 2). Tasks were individually chosen based on each child's needs. Selection of tasks involved review of recent assessments, consultation with parents and therapists, observations of treatment sessions, and informal probes to identify an area in need of intervention (e.g., conversational speech, pretend play). The authors identified two tasks that could be used to target this type of behavior.

The similarity of the tasks was independently assessed by three paraprofessionals who were blind to the purpose of the study and had at least 1 year of experience treating children with ASD. The paraprofessionals used written summaries of the relevant information (e.g., the operational definitions, included stimuli or used scripts, etc.) to complete a rubric examining the similarity of the two tasks across several domains (e.g., type of behavior, social, verbal, and motor skills required). The paraprofessionals did not identify any meaningful differences and agreed that the selected tasks were of similar difficulty. Additionally, all of the children performed identically on each of their two tasks during baseline sessions.

Tasks were randomly assigned to the video modeling via traditional technology or portable technology condition in an effort to equally distribute any differences between tasks across conditions. Once baseline sessions were completed, the tasks were written on folded paper and placed in a container. The task pulled from the container was targeted with video modeling via traditional technology on the television and the remaining task was targeted with video modeling on the iPad®.

Video Modeling Videos

A video was made for each task. Each video depicted a familiar model correctly demonstrating the targeted behaviors. Adult models were used because past research has found that adults are effective models (McCoy and Hermansen 2007) and tend to respond quickly to training and direction. Consistent with past autism research, the models looked into the camera when speaking and demonstrated speech and nonverbal actions at an exaggeratedly slow pace (e.g., Charlop and Milstein 1989). The videos shown in both conditions followed identical protocols to increase the likelihood that the

Table 2 Targeted tasks, procedures, dependent measures, and mastery criterion

Child	Behavior	Task Targeted w/ iPad®	Task Targeted w/ TV	Initial Instruction	Trial Present.	Session Length/# of Trials	Correct Response	Natural Contingency Activated by Correct Responses	Mastery Criterion (2 Consecutive Sessions)
Logan	Pretend play	Pretend play w/ <i>Star Wars</i> stimuli	Pretend play w/ <i>Batman</i> stimuli	"Play space" or "Play Batman"	None	2 min session	Pretend play verbalizations: one or more words directed towards a play character or activity that was said at an audible volume. A new verbalization began when speech was spoken through another character or after at least 3 s of silence.	None	3 or more pretend play verbalizations
Anna	Conditional mands	Mand w/ color	Mand w/ shape	None	Presenting 2 edibles that differed along the relevant dimension	6 trials	Conditional mand: verbally requesting one of the two presented edibles with a sentence including a pronoun, verb, the relevant adjective, and noun with the words in the proper order at an audible volume within 10 s of the S ^D .	The interaction partner said a 1 word response (e.g., okay) and gave her a piece of the edible	80 % of opportunities correct
Brian	Asking questions	Ask where question	Ask who question	"It's time to ask 'Where is it'" or "It's time to ask 'Who is it'"	Hiding a toy or presenting a picture	5 trials	Where/who question: verbally asking the "Where/Who is it?" or a functionally equivalent question (e.g., "Where did it go," "Who is that") at an audible volume with all words in the correct order within 10 s of the S ^D . Comprehensible word approximations were considered correct.	A two word vocal response (i.e., "It's here!" or "It's Jake!")	80 % of opportunities correct

Table 2 (continued)

Child	Behavior	Task Targeted w/ iPad®	Task Targeted w/ TV	Initial Instruction	Trial Present.	Session Length/# of Trials	Correct Response	Natural Contingency Activated by Correct Responses	Mastery Criterion (2 Consecutive Sessions)
Jason	Conversational speech	Converse about school	Converse about family	"Let's talk"	Asking the scripted question	3 trials	Conversational turn: verbally answering a question with relevant information and asking a question related to the current topic within 10 s of the S ^D . If Jason began asking a question within 10 s, the trial was extended until he finished the phrase or was silent for 2 s. All speech had to be comprehensible but word approximations and improper use of pronouns were coded correct.	The interaction partner responded to the posed question	100 % of opportunities correct
Riley	Interactive play	Play modified Old Maid	Play modified Go Fish	"Let's play" and tapping the cards he needed to pick from.	"Your turn."	5 trials	Appropriate play: completing a turn according to established rules. Both games involved selecting a card, comparing the selected card to other cards, verbally indicate the presence or absence of a match, and placing the card in the proper location. The card was selected within 10 s of the S ^D and all steps were completed within 30 s.	The play partner took her turn.	80 % of opportunities correct

only meaningful difference between conditions was the mode of delivery. To avoid any unintentional bias, all videos were created before the tasks were assigned to a condition. Videos were edited until the research team agreed they were as similar as possible. Three trained paraprofessionals with several years of experience working with children with ASD who were unaware of treatment condition independently rated video similarity. The paraprofessionals used a rubric to compare each pair of videos across numerous dimensions (e.g., length of video, number of exemplars, camera orientation). They did not identify any differences and all pairs of videos were deemed similar.

Materials

Videos assigned to the traditional technology condition were shown on a Toshiba television with a 20" screen. The television was selected because it is traditionally used in video modeling research. Videos assigned to the portable video modeling condition were played on an Apple iPad® 2. The tablet was 9.5×7.31×0.37 in. and weighed 1.44 lb.

Settings

Sessions were held in a treatment room, play room, community café, or outside. Each child's baseline, treatment, and follow-up sessions were conducted in one of these locations. Generalization probes took place in a second location (e.g., Jason's baseline, treatment, and follow-up sessions were held at the café and his generalization sessions were conducted outside).

Design and Procedure

This study used a multiple baseline design across children and an adapted alternating treatments design within child. This experimental design has been used in past autism research (e.g., Charlop, Kurtz, and Casey 1990; Taylor and Levin 1998).

The procedures for each child's two tasks were designed to be as comparable as possible to protect internal validity. Procedural similarity was confirmed by three paraprofessionals with at least 1 year of experience treating children with ASD. The paraprofessionals compared the procedures within each pair of tasks across dimensions (e.g., initial instruction, number of opportunities, length of opportunity), did not identify any meaningful differences, and agreed that the procedures within each pair of tasks were comparable. To control for possible order effects, the order in which the two tasks were presented first was counterbalanced both across children and within child across sessions. Subsequent pairs of sessions began with the task that had ended the preceding pair of sessions.

Baseline Baseline sessions assessed initial demonstration of each targeted behavior. Baseline length differed across children to control for a number of potential threats to internal validity.

The procedures used for each child depended on the targeted tasks. In general, the experimenter brought the child to the testing setting. The setting included all of the

interaction partners and materials (e.g., toys, games) needed to demonstrate the targeted behavior. The experimenter gave an initial instruction and presented opportunities to engage in the targeted behavior. Correct responses were met with natural contingencies. Incorrect or no responses were ignored and led to the termination of the trial. The specific baseline procedures for each child are briefly described in Table 2.

Generalization Generalization probes across person and setting were conducted during baseline and post-intervention. Generalization probes followed the same procedures as baseline sessions.

Intervention During intervention, testing sessions were preceded by presentation of the video modeling. The order in which the sessions occurred was alternated across children and within child across pairs of sessions. Brian and Riley began treatment with video modeling on the television and Logan, Anna, and Jason first watched video modeling on the iPad®. Each pair of sessions began with the task that had ended the preceding pair of sessions.

Video Modeling on the iPad The child watched the video on the iPad® in the testing setting. The experimenter instructed the child to sit in a chair directly in front of the screen and instructed the child to watch the video (i.e., “Watch the video”). The first time the video was presented, the child watched it twice. During subsequent presentations, the child watched the video one time. If the child exhibited off-task behavior (e.g., looked away from the screen, attempted to talk to the therapist), the therapist immediately verbally or gesturally prompted the child to pay attention. Children generally complied with the first prompt given and if they did not the prompt was repeated. No additional instruction or reinforcement was given. These procedures are consistent with past video modeling research (e.g., Charlop and Milstein 1989; Charlop-Christy, Le, and Freeman 2000). Immediately after video presentation, the therapist set the iPad® aside and began a testing session for that task. The average time between the end of the video presentation on the iPad and the start of the testing session was between 3 and 9 s across children.

Video Modeling on the Television When video modeling was presented on the television, the child watched the video on the television in the office of the treatment center. The child was instructed to sit on a chair directly in front of the screen. Otherwise, the procedure was the same as in the video modeling on the iPad® condition. Immediately after watching the video, the child was brought to the testing setting and participated in a testing session to assess his or her demonstration of the targeted behavior. The average time between the end of video presentation on the television and the beginning of the testing session for each child was 31 to 112 s.

Testing Sessions Intervention testing sessions followed the same procedure as baseline sessions with one minor modification. After presenting the initial instruction and before presenting the first trial, the therapist said “Do what you saw in the video.” When a child demonstrated the target behavior at or above criterion level during a testing session, he/she did not watch the video before the next testing session for that skill. If the child again demonstrated the behavior at or above criterion, the child was

considered to have mastered the targeted skill and intervention was discontinued. If the child did not do so, video presentation was reintroduced.

Follow-up Follow-up probes were conducted 4 to 15 weeks after a child met criterion for a task. Follow-up probes were identical to baseline probes.

Scoring

Logan's sessions were coded for the total number of pretend play verbalizations. The other five children's sessions were all composed of trials that were coded as correct or incorrect based on the previously described operational definitions.

Inter-Observer Agreement

At least 20 % of each child's sessions were scored by a second, independent coder. Inter-observer agreement for Logan's sessions was calculated by dividing the smaller number of observations by the larger number of observations and multiplying this number by 100. Inter-observer agreement of the other children's sessions was determined by dividing the number of agreements by the sum of the agreements and disagreements and multiplying this number by 100. Inter-observer agreement was 100 % across children and tasks.

Procedural Integrity

Over 33 % of each child's sessions were coded for procedural integrity. Sessions were coded using a rubric designed to measure the extent to which the video modeling intervention was implemented when and as intended and that testing session included the proper instruction, opportunity presentation, opportunity length, and established consequences. Treatment fidelity was high both across children and within child across condition. Treatment fidelity was 100 % in Logan's video modeling on the iPad® and video modeling on the television conditions, 100 % in Anna's video modeling on the iPad® condition and 98 % in the video modeling on the television condition, 99 % for Brian's video modeling on the iPad® sessions and 100 % in the video modeling on the television condition, 99 % for Jason's video modeling on the iPad® and video modeling on the television sessions, and 96 % for Riley's video modeling on the iPad® sessions and 95 % for his video modeling on the television sessions.

Social Validity of Treatment Effects

The social validity of treatment outcomes was assessed to examine the extent to which treatment gains were practically significant. Twenty therapists with experience with children who are neurotypical and diagnosed with ASD were recruited from the behavioral treatment center. Eighty-five percent of the raters were blind to the purpose of the study and all of the raters were blind to treatment condition. The raters watched 30 s video clips that were randomly selected from one of each child's baseline sessions and one of each child's final two intervention sessions for each of their two tasks. Clips

were presented in a random order and coders were not told whether clips were from baseline or post-treatment. After watching each clip, the coders independently completed a questionnaire regarding the child's behavior. The questionnaire contained 5 items presented on a 7-point Likert scale (1=Strongly Disagree to 7=Strongly Agree). Two of the items were specific to that child's tasks (e.g., Logan's were "This child demonstrates appropriate speech when playing with toys" and "This child has good pretend play skills"). The other three items were related to the child's overall social skills. One-tailed Wilcoxon signed-rank tests were used to test for statistically significant differences between ratings of each child's pre- and post-treatment behavior.

Results

Acquisition, Maintenance, and Generalization of Dependent Measures

Video Modeling on the iPad® Video modeling on the iPad effectively facilitated acquisition (see Fig. 1). Gains were generalized across person and setting and maintained 4 to 15 weeks following intervention.

Acquisition None of the children exhibited correct responses during baseline sessions. Logan met criterion after two presentations of video modeling on the iPad. Viewing video modeling on the iPad® immediately increased Anna's conditional mands to 100 % of opportunities, but her performance deteriorated when the video was not shown before her second testing session. Reintroduction of the portable video modeling intervention returned Anna's correct responses to 100 % of opportunities after 2 video modeling presentations. Brian mastered 'who' questions after 3 video modeling presentations. Jason's conversational speech began to improve after two presentations of video modeling on the iPad® and he required seven presentation of video modeling on the iPad® to meet criterion. Video modeling on the iPad® immediately improved Riley's interactive play during Old Maid sessions, but his correct responding decreased when the video was not shown immediately before at testing session. He watched video modeling on the iPad® 6 times over 8 sessions before meeting criterion.

Generalization Treatment gains were generalized across person and setting. None of the children demonstrated correct responses during pre-treatment generalization probes. During post-treatment generalization probes, all five children responded correctly at above mastery level (see Fig. 1).

Maintenance The children continued demonstrating the behavior targeted with video modeling on the iPad® during follow-up probes conducted 4 to 15 weeks following intervention. Four weeks after completing treatment, Landon said 10 and 5 pretend play verbalizations per probe. Anna, Brian, Jason, and Riley responded correctly on 100 % of follow-up opportunities 5, 7, 15, and 11 weeks after meeting criterion.

Video Modeling on the Television Video modeling delivered on the television effectively increased the children's demonstration of the targeted behavior (see Fig. 2). Four

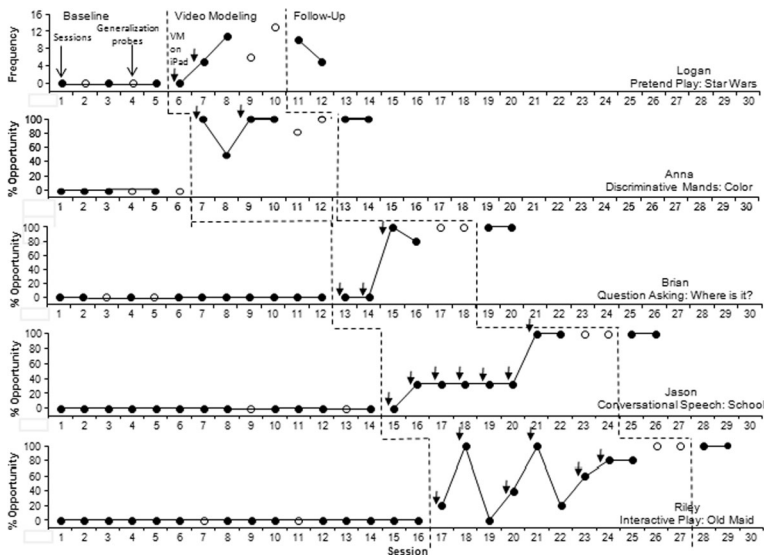


Fig. 1 Video modeling on the iPad®: Baseline, treatment, generalization, and follow-up. *Note.* Closed circles represent the child's performance during testing baseline, intervention, and follow-up testing sessions. Open circles represent the child's performance during generalization probes. Short arrows indicate that video modeling was presented on the iPad® before the testing session

of the five children generalized gains across person and setting and all continued responding correctly 4 to 15 weeks after completing the intervention.

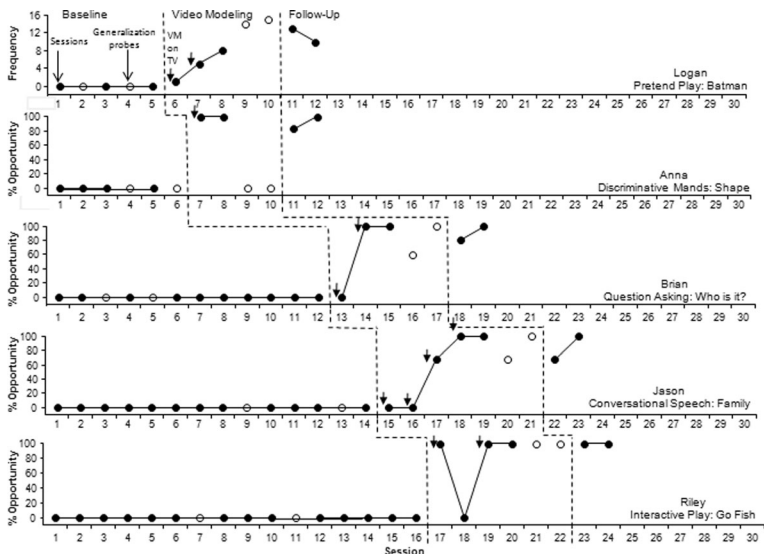


Fig. 2 Video Modeling on the television: Baseline, treatment, generalization, and follow-up. *Note.* Closed circles represent the child's performance during testing baseline, intervention, and follow-up testing sessions. Open circles represent the child's performance during generalization probes. Short arrows indicate that video modeling was presented on the television before the testing session

Acquisition Consistent with the iPad® condition, none of the children demonstrated any correct responses during baseline. As in the iPad® condition, two video modeling presentations on the television enabled Logan to meet criterion. In contrast to the iPad® condition, Anna met criterion after only 1 video modeling presentation. Brian's demonstration of 'Where' questions improved slightly faster in this condition, with two video modeling presentations on the television increasing his correct responding to 100 % across two consecutive sessions. Jason acquired the targeted conversational speech more rapidly in this condition and met criterion after only four video modeling presentations on the television. Riley mastered the modified Go Fish game after only two video modeling presentations on the television.

Generalization All but one child generalized gains across person and setting. As in the video modeling on the iPad® condition, no correct responses were observed during baseline generalization probes. Logan demonstrated even more correct responses during his post-treatment generalization probes following video modeling on the television (i.e., 14 and 15 pretend play verbalizations per probe). During post-treatment generalization probes, Brian, Jason, and Riley demonstrated marginally less generalized responding than in the iPad® condition, with correct responses on 80, 81, and 100 % of opportunities. Anna did not demonstrate any correct responses during her post-treatment generalization probes.

Maintenance As with the video modeling on the iPad® condition, gains were maintained during follow-up probes conducted 4 to 15 weeks after meeting criterion. Logan's demonstration of pretend play was slightly better following the video modeling on the television treatment, with 13 and 14 pretend play verbalizations exhibited during follow-up probes conducted 7 weeks after he met criterion. Anna, Brian, and Jason's performance during maintenance probes was robust, but slightly less than that observed in the iPad® condition. They demonstrated correct responding on 91, 90, and 83 % of opportunities. As in the iPad condition, Riley played the modified Go Fish game correctly 100 % of opportunities.

Comparison of Video Modeling on the iPad® and Video Modeling on the Television Both of the interventions resulted in behavior acquisition. Four children acquired the targeted behaviors faster when video modeling was presented on the television. Generalization and maintenance of treatment gains occurred in both treatment conditions, but tended to be slightly better in the video modeling on the iPad® condition.

Social Validity of Treatment Effects

Analyses of social validity ratings indicate that therapists noted statistically significant behavioral changes in all participants from pre- to post-treatment across treatment conditions (see Tables 3 and 4). The obtained ratings indicate that each child demonstrated noticeably improved demonstration of the targeted behavior and that the children exhibited better overall social skills after completing the intervention.

Table 3 Video modeling on the iPad®: social validity of treatment effects

	Logan			Anna			Brian			Jason			Riley		
	Mean(SD)			Z			Mean(SD)			Z			Mean(SD)		
	Pre	Post		Pre	Post		Pre	Post		Pre	Post		Pre	Post	
1. Item specific to the target behavior.	4.0(1.4)	5.1(1.4)	-3.0**	1.4(1.2)	6.5(0.8)	-4.0**	1.5(0.6)	6.5(0.8)	-4.0**	3.0(1.4)	6.3(0.7)	-3.7**	3.7(1.6)	6.8(0.4)	3.8**
2. Item specific to the target behavior.	5.5(0.9)	6.1(0.9)	-2.1**	1.4(1.3)	5.8(0.7)	-4.0**	1.7(0.6)	4.0(1.5)	-3.9**	2.2(0.9)	5.9(0.8)	-3.9**	4.1(1.3)	6.3(0.8)	-3.5**
3. This child has good social skills.	4.0(1.3)	4.6(1.1)	-2.0*	1.8(1.4)	5.0(1.1)	-4.0**	2.3(0.7)	3.8(0.9)	-4.0**	2.4(0.7)	5.4(1.0)	-4.0**	4.2(1.1)	6.1(0.8)	-3.8**
4. Other children would want to interact with this child.	4.4(1.3)	4.8(1.1)	-1.6	2.0(1.7)	5.2(1.0)	-4.0**	2.3(0.8)	4.0(0.9)	-3.5**	2.7(0.9)	5.4(0.8)	-3.9**	4.3(1.1)	6.1(0.8)	-3.8**
5. I want to interact with this child.				3.0(1.1)	5.4(1.2)	-3.6**	3.1(1.1)	4.4(1.1)	-3.2**	3.6(1.3)	5.6(1.0)	-3.4**	5.2(1.5)	6.3(1.0)	-2.9**

* $p < 0.05$ ** $p \leq 0.01$

Table 4 Video modeling on the television: social validity of treatment effects

	Logan			Anna			Brian			Jason			Riley		
	Mean(SD)			Z			Mean(SD)			Z			Mean(SD)		
	Pre	Post		Pre	Post		Pre	Post		Pre	Post		Pre	Post	
1. Item specific to the target behavior.	3.2(1.3)	5.0(1.3)	-3.3**	1.3(0.7)	6.3(0.9)	-4.0**	1.3(0.6)	6.5(0.7)	-4.0**	2.8(1.3)	4.8(1.4)	-3.3**	3.1(1.5)	6.5(0.7)	-3.9**
2. Item specific to the target behavior.	4.8(1.3)	6.1(0.8)	-3.3**	1.2(0.5)	5.2(1.5)	-4.0**	1.7(0.7)	4.4(1.7)	-3.9**	1.9(0.6)	4.2(1.4)	-3.7**	3.9(1.3)	5.6(1.2)	-3.4**
3. This child has good social skills.	3.6(1.0)	4.6(1.1)	-3.0**	1.6(0.7)	4.7(1.1)	-4.0**	1.9(0.6)	4.2(1.1)	-3.8**	2.2(0.7)	3.5(1.2)	-3.4**	3.7(1.2)	5.0(1.2)	-3.0**
4. Other children would want to interact with this child.	3.7(1.2)	4.8(1.4)	-2.5*	1.9(0.7)	4.8(1.2)	-3.9**	2.5(1.1)	4.3(1.0)	-3.8**	2.6(0.8)	3.9(1.2)	-3.1**	4.3(0.9)	5.6(1.3)	-3.0**
5. I want to interact with this child.	4.5(1.4)	5.5(1.4)	-3.0**	3.0(1.3)	5.2(1.0)	-3.6**	3.3(1.3)	4.8(0.9)	-3.1**	3.2(1.2)	4.2(1.1)	-2.4*	5.1(1.2)	6.4(0.9)	-3.2**

* $p < 0.05$ ** $p \leq 0.01$

Further, the therapists rated the children as more attractive interaction partners during post-intervention clips.

Discussion

Recent technological advances have produced a range of portable handheld devices that are being used to deliver video modeling treatments to children with ASD. Video modeling on the iPad® and video modeling on the television both led to mastery of target behaviors. However, video modeling on the iPad® resulted in slightly slower acquisition of the target behaviors for four participants. These children required 1 to 4 more video modeling presentations when treatment was implemented on the iPad®. This should be considered when selecting a treatment modality.

The reason for the obtained difference in rate of acquisition is unclear. Past research suggests that at least some adults with developmental disabilities prefer larger screens (Carey, Friedman, and Bryen 2005). More specific to children with ASD, research has found that stimuli size can affect learning. For example, research on within stimulus prompts has found that enlarging the size of one aspect of stimuli relative to the rest of the item enables children to discriminate between stimuli along this dimension (Schreibman 1975). It is possible that the presentation of larger stimuli, in general, may facilitate the learning of children with ASD. Another possibility is that use of an assumedly preferred device affects acquisition. The incorporation of highly preferred items has been found to effectively promote acquisition (e.g., Charlop-Christy and Carpenter 2000; Ingersoll and Schreibman 2006), but these strategies use access to preferred items to reinforce the targeted behaviors. In contrast, the iPad® was only used to administer the treatment and the children were not given the opportunity to use the device for more preferred activities. Involving preferred stimuli but not using it in the desired manner (i.e., video games) may have a different effect on treatment effectiveness. Additional research examining these possible explanations is needed.

Generalization and maintenance were high following both treatment conditions. Four children demonstrated different amounts of generalization and maintenance across treatment conditions and most of the obtained differences favored the video modeling interventions shown on the iPad®. During generalization and maintenance probes, Anna, Brian, and Jason demonstrated better generalization of the behaviors targeted with video modeling on the iPad®. This may have resulted from the iPad's® greater prevalence across the children's natural settings, the increased common stimuli present during delivery of the portable video modeling treatment, or both (Charlop et al., 2002; Petursdottir et al. 2007; Stokes and Baer 1977). An alternative possibility is that the better generalization data were due to the additional number of treatment sessions these children had due to their need to meet acquisition criterion with the iPad®. That is, all of the children who demonstrated better generalization and maintenance following video modeling on the iPad® participated in more treatment sessions in this condition. This additional exposure over a longer period of time may have produced more fluent responding that led to the observed differences (Green 2001; Kubina and Wolfe 2005). The finding that Logan, the only child who watched the same number of video modeling presentations on both mediums, demonstrated better generalization and maintenance following video modeling on the television may support this theory.

The mixed findings across outcome variables make it difficult to reach a definitive conclusion on the superiority of one mode of implementation. While acquisition, generalization, and maintenance are all widely recognized as critical measures of treatment effectiveness, there is no clear delineation of effect sizes needed to indicate meaningful differences or the weight that should be assigned to each factor when evaluating an intervention (Cooper et al. 2007; Reichow et al. 2008; Stokes and Baer 1977). However, the present data seem to slightly favor the use of video modeling on the television over the iPad®. Differences in rate of acquisition consistently favored the television monitor. Over time, the cumulative effects of this faster intervention could allow children to achieve substantially better functioning. In contrast, the iPad® intervention was only associated with better generalization and maintenance of acquired skills by three of the children and the generally small differences may not lead to meaningfully improved treatment outcomes. Even when lower, the generally high generalization and maintenance following video modeling on the television are likely also sufficient in enabling children to contact the existing natural contingencies and learning opportunities across settings that would continue facilitating their skill development.

While video modeling on traditional technology may be slightly more efficient, the present data indicate that video modeling on the iPad® does effectively promote positive outcomes in children with ASD. It is worth noting that Brian and Jason demonstrated behavior change after watching the videos in more distracting environments with auditory and visual stimuli that were not under experimental control. This is significant because ease of implementation across such environments has been cited as a potential benefit of portable video modeling interventions, but past studies that have successfully used video modeling interventions on portable handheld technologies in these more distracting environments have also included other prompting procedures (e.g., Ayres et al. 2013; Carnahan et al. 2012; Cihak et al. 2010; Hart and Whalon 2012). Additionally, behavioral changes were socially validated, indicating that observed changes were great enough to have practical effects (Kazdin 2005).

Many steps were taken to strengthen internal validity, but there were limitations. A staggered baseline within participants across behaviors would have enhanced the design. The utilized random assignment to treatment condition may not have successfully distributed differences due to the small sample size. It is also important to note that the present findings on the comparative and general effectiveness of video modeling interventions delivered via the iPad® only apply to video modeling interventions that were individually created for each child and follow empirically validated treatment procedures.

The present study provides preliminary evidence that delivering video modeling interventions on portable handheld devices does not increase the effectiveness of the intervention. The most reliable difference obtained in the present study suggests that using the portable handheld technology may lead to slightly slower acquisition. This finding highlights the importance of empirically examining all changes to intervention strategies to provide information that guides practitioners in selecting the interventions that best facilitate child progress. Conducting additional research examining the comparative effectiveness of the use of different technologies may be an especially

important area of research due to the already widespread use of portable handheld devices in autism treatments.

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